Imbalance & stuff

Putting a Number on the Risk You're Already Taking

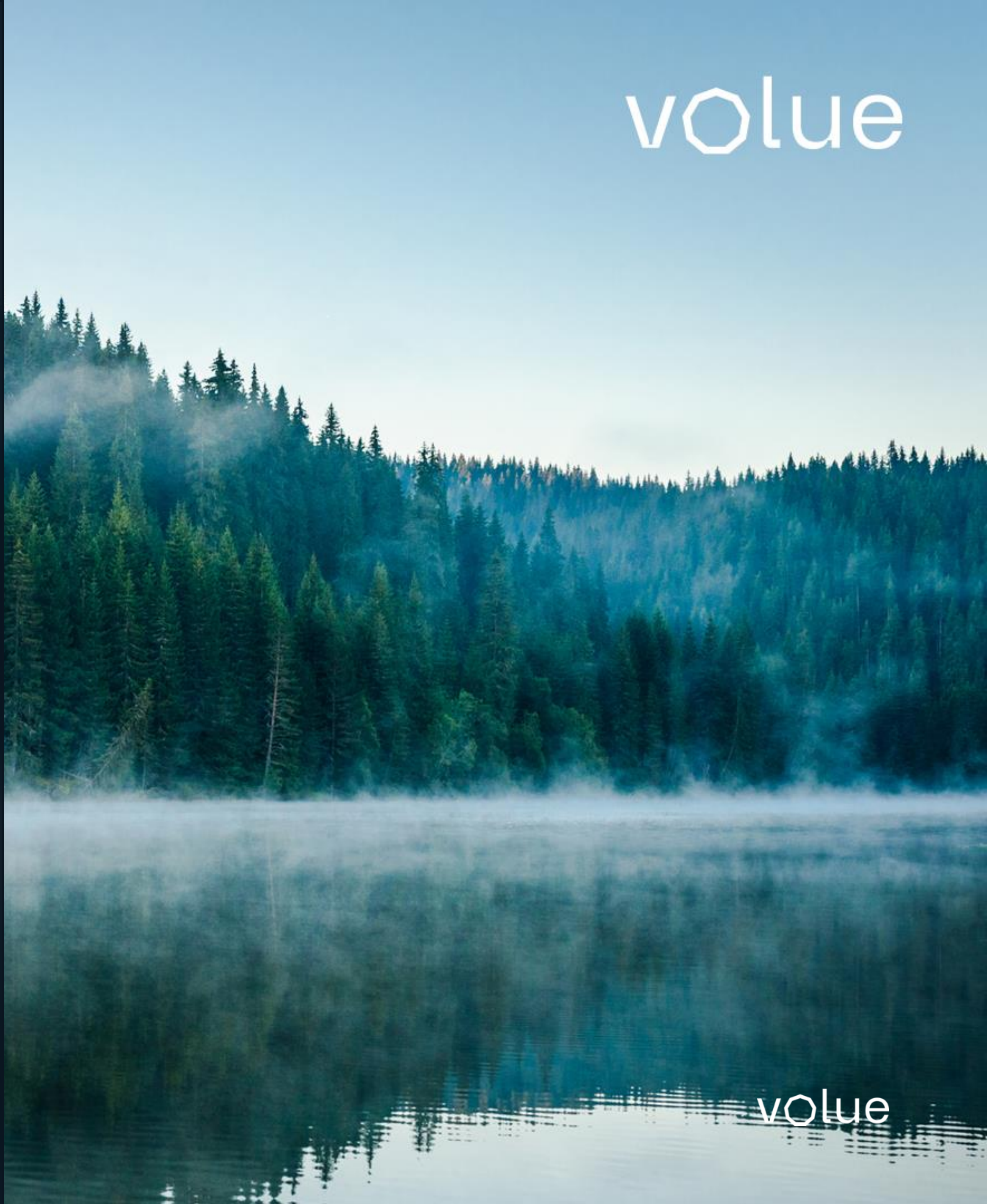
Power Risk Management · Nordic & European Markets · 2026

Agenda

Total: 50 mins

FCR market changes

Starting/started today!



FCR markets in the Nordics

Mainly Sweden, Denmark and Norway, to a lesser extent Finland

Both auctions using the same algorithm

Previously, only the 2nd (D-1) FCR auction utilized the Optimeering clearing algorithms

As of today, this automated approach will be applied to both the 1st and 2nd auctions

Transition from Manual Selection: The 1st auction has moved away from manual price-order selection. While this may lead to an increase in "skipped bids," it is expected to lower TSO payouts and market prices overall

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Further, the TSO's will limit how much static volume is allowed to cover as % of the demand

Given everything else equal – prices will increase if the maximum static limit is low and binding

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Complex bids

Divisibility: Users can now submit divisible bids

Fixed Block Bids: Block bids have transitioned from floating to fixed lengths.

Introduction of divisibility may help reduce prices and the frequency of skipped bids

This shift toward rigid block lengths may increase prices and the likelihood of bids being skipped

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Imbalance at Risk

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Imbalance Costs Are Material — And Growing

Imbalance settlement is no longer a rounding error. For renewable-heavy portfolios it is one of the largest variable cost lines on the P&L.

5–10%

of monthly settlement revenue at risk

For an unmanaged wind portfolio in a volatile month, imbalance settlement costs can approach or exceed 10% of energy revenue.

3–15×

price spike multiplier in extreme events

Nordic and Central European imbalance prices can spike 3–15× above day-ahead during weather-driven supply shocks.

96

separate settlement intervals per day

15-minute MTUs mean 96 independent exposure windows per day — each requiring its own risk assessment.

These figures are indicative. Precise exposure varies by portfolio, area, and market conditions.

We Measure Everything Else

Power market risk management is sophisticated. But one risk is systematically under-measured.

Risks we already quantify

- ✓ Price risk — mark-to-market, VaR, Greeks
- ✓ Volume risk — production forecasts, tolerance bands
- ✓ Credit risk — counterparty limits, CVA
- ✓ Basis risk — area price spreads, cross-hedging
- ✓ Regulatory risk — compliance frameworks, stress tests

Imbalance settlement risk

?

No standard metric. No confidence interval.
No formal limit framework.
Managed by intuition, if at all.

The Same Position Can Mean Very Different Costs

A position report tells you what you hold. It tells you nothing about what settling it will cost.

Scenario A — Favourable

Position: 50 MWh net short

Imbalance price: €9/MWh

System: long (excess renewable)

Settlement cost: €450

Scenario B — Adverse

Position: 50 MWh net short

Imbalance price: €94/MWh

System: short (low wind, high demand)

Settlement cost: €4,700

10× difference in cost from the same position. Risk management requires a probability framework— not just a position report.

Introducing Imbalance at Risk (IaR)

"With 95% confidence, our imbalance settlement cost over the next 24 hours will not exceed

€X"

IaR is to imbalance settlement what Value at Risk (VaR) is to a trading portfolio — a single, probabilistic, confidence-interval-bounded risk measure.

Gross IaR

Measures the worst-case total imbalance settlement cost: position × imbalance price.

Captures the absolute cash flow risk — the settlement bill you will pay (or receive).

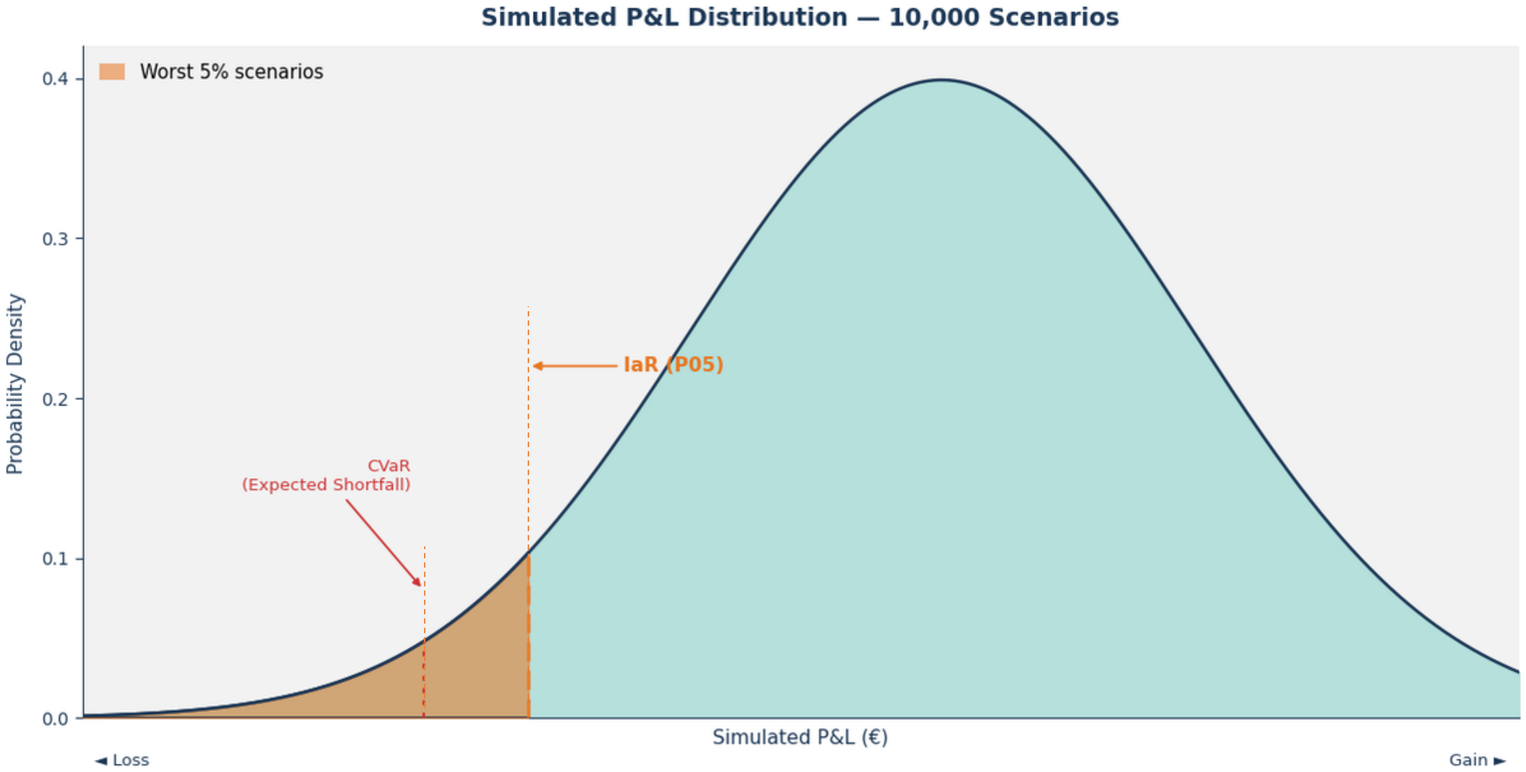
Spread IaR

Measures the worst-case underperformance vs the day-ahead market: position × (imbalance price − DAM price).

Captures economic performance risk — how much worse you did vs a perfect hedge.

IaR Is a Point on a Distribution — Not a Single Estimate

Running thousands of scenarios produces a full P&L distribution. IaR is simply the loss at the chosen confidence level.



IaR (P05)

The threshold at which only 5% of simulated scenarios show a larger loss. Your 'risk budget' for settlement exposure.

CIaR / Expected Shortfall

The average loss across the worst 5% of scenarios. Tells you what happens when you breach the IaR — how bad can it get?

The shape matters. A portfolio with IaR €40k but heavy tail risk is more dangerous than one with IaR €50k and thin tails.

IaR Works Across All Relevant Time Horizons

The same framework scales from real-time operational monitoring to weekly portfolio governance.

1

Per-MTU (15 minutes)

Real-time

IaR for each individual settlement interval. Identifies which MTUs are driving the most risk right now. Powers real-time alerts and trading signals.

Users: Intraday operators, dispatch / scheduling, traders

2

Rolling Window (e.g. 1h / 4h)

Intraday

Cumulative risk over a rolling horizon. Catches risk that builds gradually across consecutive MTUs — not visible at the single-MTU level.

Users: Intraday traders, portfolio managers, intraday risk managers

3

Remaining Day / Week

Portfolio

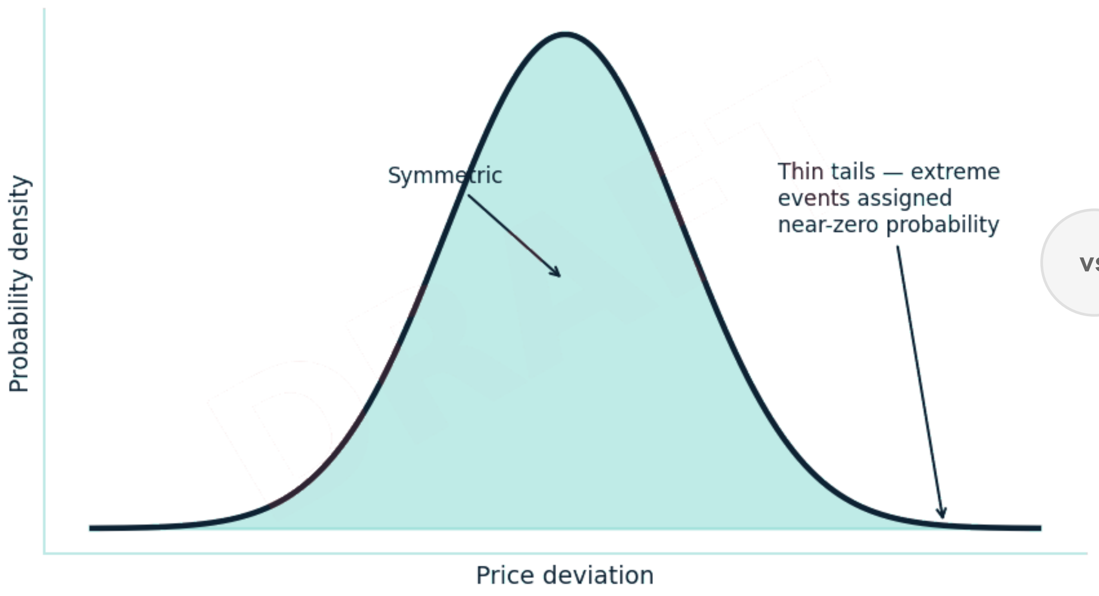
Period IaR across all remaining settlement intervals. The primary metric for limit governance and daily risk reporting to management.

Users: Portfolio managers, risk controllers, CFOs, risk committees

Challenge 1: Imbalance Prices Don't Follow a Normal Distribution

Standard statistical models systematically underestimate tail risk. The consequences of getting this wrong are not symmetric.

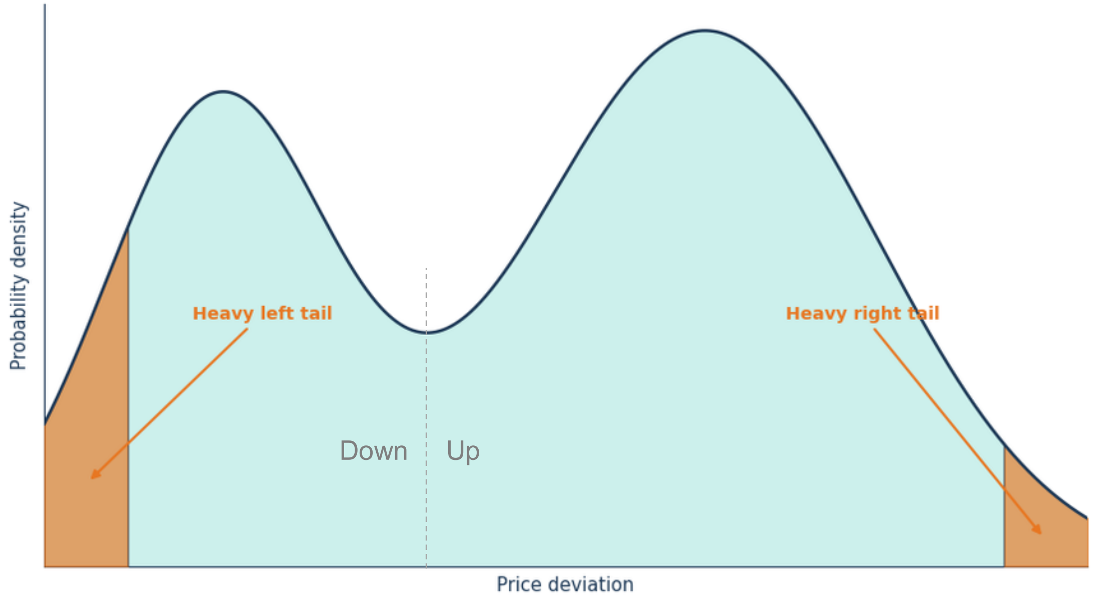
What standard models assume



Symmetric bell curve. Extreme events rare by construction. Tails thin.

vs

What markets actually look like



Asymmetric. Heavy left tail. Spike events that normal models assign near-zero probability to.

A model calibrated on normal price assumptions will understate IaR. In extreme market conditions, the actual loss may be 2–3× the modelled figure.

Challenge 2: Position and Price Are Not Independent

For wind portfolios, the worst positions coincide with the worst prices. Models that ignore this can dramatically understate risk.

1 Wind drops

Renewable generation falls below forecast.
Portfolio is now net SHORT.

→

2 System price rises

Lower renewable generation pushes up the system imbalance price.
Spike risk is highest.

→

3 Loss amplified

Short position × high price = maximum settlement cost.
Both effects compound simultaneously.

Independent model assumes:

Price and position vary randomly and separately. Estimated IaR: €28,000

Dependence-aware model shows:

Wind shortfall and high price occur together. True IaR: €48,000 (+71% higher)

Challenge 3: Risk Across MTUs Cannot Simply Be Summed

The period IaR is not the sum of per-MTU IaRs. Getting this wrong leads to systematic over-hedging or under-hedging.

Naive approach

Sum each MTU's IaR independently, then add them up.

IaR(MTU 1) + IaR(MTU 2) + … + IaR(MTU 96)

= €65,000

Assumes every MTU has its worst outcome simultaneously — an impossibility.

Correct approach

Simulate the full joint distribution of all MTUs simultaneously — then take the period quantile.

$Q_{0.05}(\Sigma \text{ P\&L across all MTUs})$

= €42,800

Requires scenarios that correctly reproduce the autocorrelation of prices across MTUs and the co-dependence between imbalance price and volume.

Diversification ratio: €42,800 ÷ €65,000 = 0.66 — 34% of 'risk' disappears when modelled correctly. The ratio is a direct measure of how much the portfolio benefits from temporal diversification.

Challenge 4: IaR Must Respond in Real Time, on Every Event

An IaR figure computed at 08:00 is stale by 10:00. This is not a daily reporting problem — it is a continuous data and compute problem.

96

MTUs per day

Each a separate random variable. A 24h horizon requires a 96-dimensional joint simulation. Extending to 48h means 192 dimensions.

5,000+

scenarios per cycle

Enough Monte Carlo samples to reliably estimate the 5th percentile. Fewer scenarios = noisy IaR estimates that trigger false alerts or miss real risk.

Event-driven

recomputation cycle

IaR is recalculated whenever a key driver changes — a new Optimeering forecast, a position update, or a new intraday price signal. Triggered by events, not a clock.

3 inputs

must synchronise live

Price forecast marginals (Optimeering), position data (portfolio system), and historical copula calibration (model layer) — all updated, validated, and aligned.

This is the combination of quantitative finance, forecast integration, and real-time engineering that makes IaR genuinely hard to build well.

Use Case: Intraday Risk Management

IaR transforms settlement management from reactive (post-settlement analysis) to proactive (pre-settlement risk control).

Without IaR	With IaR
— Operator sees position report — 55 MWh long at 14:00	✓ Live IaR dashboard: €42,800 at 85% of limit
— No sense of what this means for settlement cost	✓ Colour-coded heatmap shows risk by MTU and hour
— Limits set as fixed volume thresholds (e.g. 'max 80 MWh')	✓ Configurable limits: per-MTU, rolling 4h, remaining day
— Imbalance cost only known at settlement — hours later	✓ Alerts fire before limits are breached — time to act
— Risk management is reactive: post-event analysis	✓ Risk management is proactive: pre-settlement decisions

Key question answered: "Do I need to trade now to reduce my risk, or is my current position within acceptable bounds?"

Use Case: Intraday Trading Decisions

Every intraday trade changes the portfolio's IaR. This creates a framework for pricing the risk-reduction value of a trade.

Marginal IaR: the change in period IaR per unit of position changed — the 'risk price' of each MWh of exposure.

1

Identify the risk driver

Marginal IaR reveals which MTUs are contributing most to period risk. MTU 15:00–15:15 drives €2,140 of the €42,800 total — disproportionate to its size.

→

2

Quantify the trade-off

Buying 30 MWh at 15:00 in the intraday market reduces period IaR from €42,800 to €36,400 — a risk reduction of €6,400 at a trade cost of €480.

→

3

Make the decision

Risk reduction value €6,400 >> trade cost €480. The hedge is economical. Without IaR, this comparison cannot be made — the decision defaults to intuition.

Without IaR

Traders ask: “Should I hedge?” and answer based on position size and gut feel.

With IaR

Traders ask: “What is the risk-reduction value of this trade versus its cost?” — and get a number.

Use Case: Risk Governance and Reporting

IaR gives risk committees and regulators the same framework they already use for market risk — applied to imbalance settlement.

Consistent limits

Set limits in €, not MWh. Compare risk across portfolios with different technologies, sizes, and price exposures on a single scale.

Limits that are defensible to a board because they are based on probability, not intuition.

Auditable exceedances

Every limit breach is logged, dated, and explainable. The historical view shows whether the IaR model is well-calibrated (Kupiec test) and what caused each exceedance event.

Regulatory readiness

As European energy markets tighten accountability frameworks, IaR provides a documented, consistent methodology for demonstrating imbalance risk oversight to regulators.

Without IaR (today)	With IaR (target)
Volume limits (MWh-based) — not risk-adjusted	Probabilistic limits (€-based at stated confidence level)
Different portfolios incomparable	Comparable across all portfolios and technologies
Exceedances explained post-hoc, narratively	Exceedances flagged in real-time with quantified impact

↑

Renewable intermittency

Higher wind and solar penetration means larger and more frequent imbalance events — the problem grows with the energy transition.

↑

Settlement regime tightening

Nordic and EU imbalance settlement regimes continue to evolve toward shorter intervals and tighter price signals. Exposure is increasing.

↑

Competitive pressure

Portfolios that formally manage imbalance risk can take on more exposure with confidence — and outperform those that manage by gut feel.

What would you do differently tomorrow if you had this number today?

What could your team achieve with real-time risk visibility?

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Does This Resonate? We'd Love to Hear Your View.

IaR is a concept we're exploring. Is it on your radar? Does it solve a real problem on your roadmap?

1

Tell us about your setup

Your portfolio and how you currently think about imbalance exposure. We want to understand your world before we say anything.

2

How can IaR work for you?

Let's discuss what IaR could look like for your specific context, and how it addresses your imbalance management needs.

3

We'd love your feedback

Does IaR solve a problem for you? Would you be interested in helping us to take this forward? Your feedback - positive and negative - is essential.

Find one of the Optimeering team at the Forum today.

Let's talk more about IaR together